# RIVERSIDE ANALYTICAL

Environmental Testing Laboratory
Coon Rapids, Minnesota · riversideanalytical.com

# FINAL REPORT

Certificate of Analysis
MNELAP lab ID 027-003-418

REPORT TO

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|---|---|
| Work order | **WO-26-10418** |
| Project | **Salt in the Lakes** |
| Quote | **Q-26-0931** |
| Samples received | **10/12/2026 09:15** |
| Report date | **10/20/2026** |
| Report status | **Final** |
| Project manager | **Priya Raman** |

| TEMPERATURE ON RECEIPT | CUSTODY SEAL | CONTAINERS | HOLDING TIME |
|---|---|---|---|
| 3.8 °C, cooler on ice | Intact | 4 x 250 mL HDPE, none broken | Met for all samples |

## ANALYTICAL RESULTS

Chloride by EPA 300.0, ion chromatography

| Lab ID | Client sample ID | Collected | Analyte | Result | Units | RL | DF | Analyzed |
|---|---|---|---|---|---|---|---|---|
| 10418-01 | **COR-1** | 10/10/2026 09:12 | Chloride | **208** | mg/L | 10 | 10 | 10/16/2026 |
| 10418-02 | **PEN-OUT** | 10/10/2026 10:33 | Chloride | **196** | mg/L | 10 | 10 | 10/16/2026 |
| 10418-03 | **ODG-IN** | 10/10/2026 12:20 | Chloride | **252** | mg/L | 10 | 10 | 10/16/2026 |
| 10418-04 | **SUM-1** | 10/10/2026 13:41 | Chloride | **117** | mg/L | 10 | 10 | 10/16/2026 |

Matrix: water, grab samples in 250 mL HDPE bottles. Results are on an as-received basis. Client sample IDs and collection times are as written on the chain-of-custody record. RL: reporting limit, adjusted for dilution. DF: dilution factor.
Preparation: 0.45 µm membrane filtration, then a 1:10 dilution with deionized water, on 10/16/2026. Analytical batch IC-261016-2.

## CASE NARRATIVE

1. **Sample receipt.** Four water samples were received on 10/12/2026 at 09:15 in one cooler with ice. The cooler read 3.8 °C on receipt and the custody seal was intact. All four 250 mL HDPE bottles arrived unbroken, and the sample IDs, dates and times on the labels matched the chain-of-custody record.

2. **Analysis.** Chloride was determined by EPA 300.0 on 10/16/2026, six days after collection and within the 28-day holding time. Each sample was filtered through a 0.45 µm membrane and diluted 1:10 with deionized water before injection. All four results fall inside the calibrated range of 10 to 500 mg/L at this dilution, so no sample needed a further dilution or a rerun.

3. **Quality control.** The initial calibration, calibration checks, method blank, laboratory control sample, matrix spike and sample duplicate for batch IC-261016-2 were all within acceptance limits. No results are qualified; QC details are on page 2.

4. **Chain of custody.** Line 2 of the chain-of-custody record (second collector) shows an initial only. QA has asked the client to complete that entry before the work order is closed. This does not affect the analysis or the results in this report.

## QUALITY CONTROL SUMMARY

Batch IC-261016-2 · chloride, EPA 300.0 · analyzed 10/16/2026

| QC sample | Lab ID | Spike added | Result | Recovery / RPD | Limits | Status |
|---|---|---|---|---|---|---|
| **Method blank** | MB-261016 | | <0.50 mg/L | | Below 1.0 mg/L | **Pass** |
| **Laboratory control sample** | LCS-261016 | 25.0 mg/L | 25.3 mg/L | **101%** | 90 to 110% | **Pass** |
| **Matrix spike, COR-1**<br>Unspiked result 207.8 mg/L | 10418-01 MS | 100 mg/L | 305.8 mg/L | **98%** | 80 to 120% | **Pass** |
| **Sample duplicate, COR-1**<br>Original result 207.8 mg/L | 10418-01 DUP | | 212.2 mg/L | **RPD 2.1%** | 10% or less | **Pass** |

Method blank and laboratory control sample values are as analyzed. Matrix spike and duplicate values are on the original-sample basis, with the 1:10 dilution applied. Recovery and RPD are calculated from unrounded results.

## CALIBRATION AND METHOD

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|---|---|
| **Initial calibration** | Five standards at 1.0, 5.0, 10, 25 and 50 mg/L as injected, linear fit, r = 0.9997 (limit: 0.995 or better). Pass. |
| **Calibration checks** | 10 mg/L standard at the start of the run, after 10 injections and at the end: 99%, 102% and 98% recovery (limits 90 to 110%). Pass. |
| **Calibrated range** | 1.0 to 50 mg/L as injected, equal to 10 to 500 mg/L in the original sample at 1:10. Results are not extrapolated outside this range. |
| **Reference method** | EPA 300.0, Determination of Inorganic Anions by Ion Chromatography |
| **Instrument** | Ion chromatograph with an anion-exchange column, carbonate eluent, suppressor and conductivity detector |
| **Holding time** | 28 days from collection, with samples kept at 0 to 6 °C |
| **Accreditation** | Accredited by the Minnesota Department of Health (MNELAP), lab ID 027-003-418, for chloride in non-potable water by EPA 300.0 |

## DEFINITIONS

| | |
|---|---|
| **RL** | Reporting limit, adjusted for dilution |
| **DF** | Dilution factor |
| **RPD** | Relative percent difference |
| **<** | Less than the value shown |
| **mg/L** | Milligrams per liter |
| **MB** | Method blank |
| **LCS** | Laboratory control sample |
| **MS** | Matrix spike |
| **DUP** | Sample duplicate |
| **r** | Correlation coefficient of the calibration |

## CONDITIONS

Results relate only to the samples tested, as received. This report may be reproduced only in full, unless Riverside Analytical approves otherwise in writing. The report has 2 pages; the chain-of-custody record for this work order is provided as a separate document.

REVIEWED AND APPROVED



**Priya Raman, Project Manager**
Approved 10/20/2026

QUESTIONS ABOUT THIS REPORT

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Please refer to work order WO-26-10418.

*End of report*